

DEPARTMENT OF TEXTILE TECHNOLOGY

Technical Report

Soil Moisture Sensor & Electrical Actuator

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PART I: SOIL MOISTURE SENSOR

1. Introduction

A soil moisture sensor is an electronic device used to measure the volumetric water content in soil. It plays a critical role in precision agriculture, automated irrigation systems, environmental monitoring, and smart farming applications. By providing real-time data on soil conditions, these sensors help conserve water, improve crop yield, and enable intelligent control systems.

Soil moisture sensors are widely used in greenhouses, crop fields, research stations, and smart home gardens. They interface easily with microcontrollers such as Arduino, Raspberry Pi, and other embedded systems, making them accessible and versatile.

2. Sensor Module Image

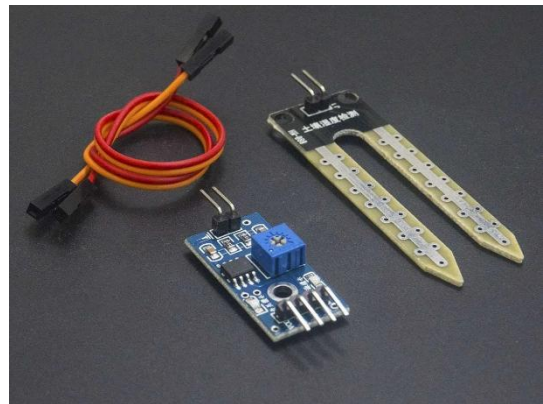
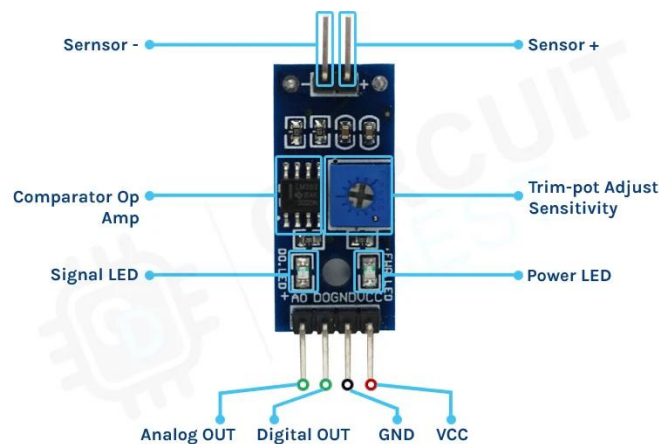


Fig 1: Soil Moisture Sensor Module

3. Working Principle

3.1 Resistive Soil Moisture Sensor

Resistive sensors work on the principle of electrical resistance. Two conductive probes are inserted into the soil, and a voltage is applied across them. Wet soil has lower electrical resistance (more ions conduct electricity), while dry soil has higher resistance. The sensor measures the resistance and converts it to a proportional analog voltage output.

3.2 Capacitive Soil Moisture Sensor

Capacitive sensors measure the dielectric constant of the surrounding soil. Water has a very high dielectric constant (~ 80) compared to air (~ 1) and dry soil ($\sim 3-5$). The sensor acts as a capacitor,

and the measured capacitance changes based on water content. Capacitive sensors are more accurate, durable, and do not corrode like resistive types.

4. Circuit Diagram

Fig 2: Soil Moisture Sensor - Wiring Diagram

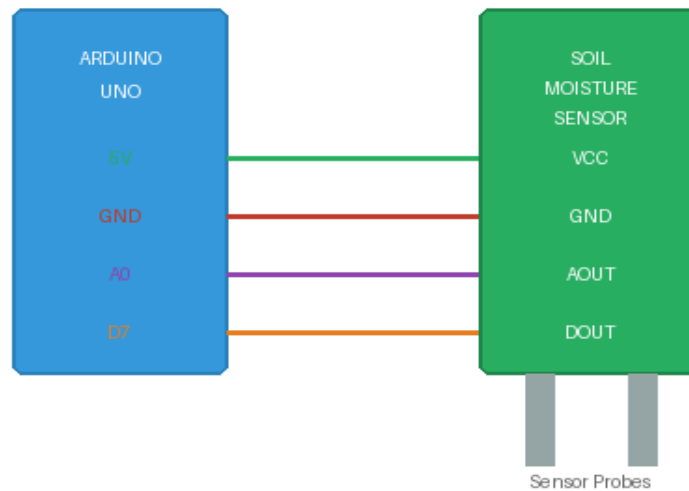


Fig 2: Circuit Diagram — Soil Moisture Sensor with Arduino

4.1 Pin Configuration

Parameter	Specification
VCC	Connect to 5V (Arduino) or 3.3V
GND	Connect to Ground (GND)
AOUT (AO)	Analog Output — variable voltage (0–5V) based on moisture level
DOUT (DO)	Digital Output — HIGH (dry) or LOW (wet), threshold set by onboard potentiometer

4.2 Arduino Sample Code

```
int sensorPin = A0;
int sensorValue = 0;
void setup() { Serial.begin(9600); }
void loop() {
  sensorValue = analogRead(sensorPin);
  Serial.println(sensorValue);
  delay(1000);
}
```

5. Technical Specifications

Parameter	Specification
Module Type	Resistive / Capacitive (YL-69, FC-28, v1.2)
Operating Voltage	3.3V – 5V DC
Output Type	Analog (0–1023 ADC) and Digital (0 or 1)
Output Voltage (Analog)	0V (Wet) – 5V (Dry)
Comparator IC	LM393 Dual Voltage Comparator
PCB Dimensions	30 mm × 16 mm (Module Board)
Probe Length	60 mm
Moisture Range	0% – 100% Relative Moisture
Response Time	< 1 second
Operating Temperature	-10°C to +85°C
Threshold Adjustment	Onboard potentiometer for DO sensitivity
Communication	Analog (AOUT) / Digital (DOUT)
LED Indicators	Power LED + Status LED (DO active)
Current Consumption	~35 mA @ 5V

6. Applications

- Automated drip irrigation and smart watering systems
- Greenhouse environmental monitoring and control
- Precision agriculture and crop management
- Soil research and environmental science experiments
- Smart home garden automation
- IoT-based remote monitoring with cloud connectivity
- Hydroponic and vertical farm control systems

7. Advantages and Disadvantages

7.1 Advantages

- Low cost and widely available (especially resistive type)
- Simple integration with Arduino, Raspberry Pi, ESP8266/ESP32
- Real-time monitoring and fast response
- Both analog and digital output options
- Capacitive variant is highly accurate and corrosion-free

7.2 Disadvantages

- Resistive sensors are prone to corrosion and electrolysis over time

- Accuracy may vary with different soil types and temperatures
- Limited depth measurement (surface/shallow measurement only)
- Requires calibration for precise volumetric readings

PART II: ELECTRICAL ACTUATOR

8. Introduction

An electrical actuator is an electromechanical device that converts electrical energy into mechanical motion. It serves as the output device in a control system, receiving a command signal from a controller and producing physical movement — such as linear displacement, rotational torque, or force — to control a process or mechanism.

Electrical actuators are used extensively in robotics, industrial automation, medical devices, HVAC systems, and agricultural machinery. Common types include solenoid actuators, DC motor-driven linear actuators, servo actuators, and stepper motor actuators.

9. Actuator Module Image

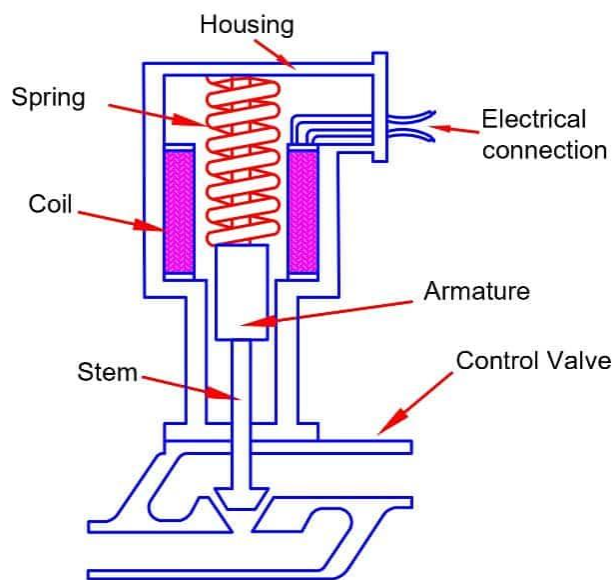


Fig 3: Electrical Actuator (12V DC Linear Motor Type)

10. Working Principle

10.1 Linear Electric Actuator

A linear actuator converts rotary motion from an internal DC motor into linear (straight-line) movement via a lead screw mechanism. When voltage is applied, the motor rotates a threaded screw, causing a nut attached to the output rod to move in or out. Reversing polarity reverses the direction of movement.

10.2 Solenoid Actuator

A solenoid actuator uses electromagnetic principles. When current flows through a coil wound around a ferromagnetic core, a magnetic field is generated. This field attracts or repels a movable iron plunger, creating linear motion. The stroke is typically short (a few millimeters) but with high force.

10.3 Servo Actuator

Servo actuators use a DC motor with a feedback potentiometer and control circuitry to achieve precise angular or linear positioning. A PWM (Pulse Width Modulation) signal controls the position, making servo actuators ideal for robotic arms, camera gimbals, and throttle control systems.

11. Circuit Diagram

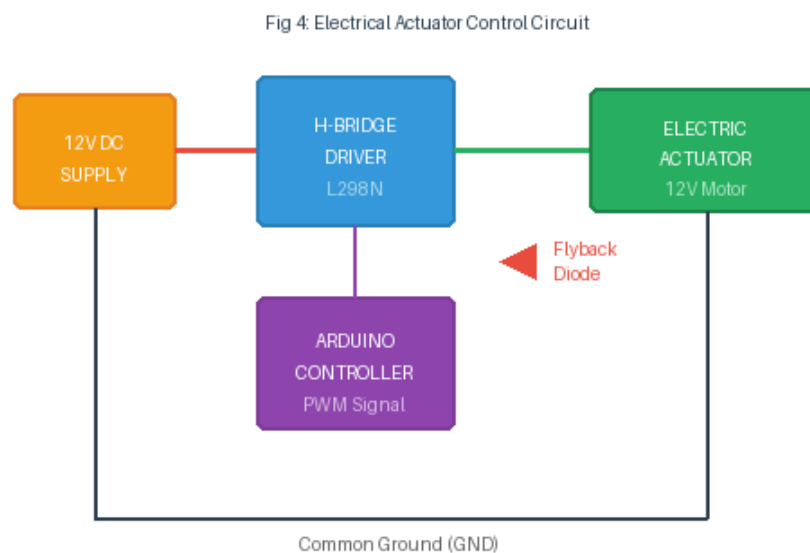


Fig 4: Electrical Actuator Control Circuit with L298N Driver

11.1 L298N H-Bridge Motor Driver Pin Description

Parameter	Specification
IN1 / IN2	Direction control inputs from Arduino (HIGH/LOW logic)
ENA	Enable pin — PWM input for speed control
OUT1 / OUT2	Motor output terminals connected to actuator motor
VCC (12V)	External power supply input for the motor
5V	Logic power supply (from Arduino or onboard regulator)
GND	Common ground for all components

11.2 Arduino Control Code

```
int IN1 = 7, IN2 = 8, ENA = 9;
void setup() {
```

```

    pinMode(IN1, OUTPUT); pinMode(IN2, OUTPUT);
    pinMode(ENA, OUTPUT);
}
void moveForward() {
    digitalWrite(IN1, HIGH); digitalWrite(IN2, LOW);
    analogWrite(ENA, 200); // Speed 0-255
}
void moveBackward() {
    digitalWrite(IN1, LOW); digitalWrite(IN2, HIGH);
    analogWrite(ENA, 200);
}

```

12. Technical Specifications

Parameter	Specification
Actuator Type	Linear DC Electric Actuator
Operating Voltage	12V DC (6V – 24V range available)
Current Draw (No Load)	~150 mA
Current Draw (Full Load)	~3 A (varies by model)
Stroke Length	100 mm – 500 mm (configurable)
Force Capacity	500 N – 2000 N (model dependent)
Speed (No Load)	5 – 30 mm/s
Duty Cycle	Up to 25% (intermittent operation recommended)
Protection Class	IP54 – IP65 (dust and splash resistant)
Motor Driver	L298N H-Bridge (up to 2A per channel)
Control Signal	PWM / Digital High-Low
Limit Switches	Built-in end-of-stroke limit switches
Operating Temperature	-20°C to +65°C
Material	Aluminum housing with plastic/metal end caps
Feedback (Optional)	Potentiometer / Hall-effect encoder

13. Applications

- Automated irrigation valve control (paired with soil moisture sensor)
- Industrial machine tool actuation and precision positioning
- Medical devices: patient beds, dental chairs, rehabilitation equipment
- Robotics: joint actuation, gripper mechanisms, mobile platforms
- Automotive: tailgate lifts, convertible roof mechanisms
- Agriculture: seed planting depth control, harvesting mechanisms
- HVAC: damper control and duct valve actuation

- Solar panel tracking systems

14. Advantages and Disadvantages

14.1 Advantages

- Clean energy conversion — no hydraulic fluid or pneumatic lines needed
- High precision and repeatability in positioning
- Easily interfaced with microcontrollers and PLCs
- Variable speed and force control via PWM
- Low maintenance with long service life
- Compact and lightweight design

14.2 Disadvantages

- Lower force output compared to hydraulic actuators
- Generates heat at high duty cycles — requires cooling
- Motor can be damaged by sustained stall current
- Requires flyback diode protection to prevent back-EMF damage

PART III: INTEGRATED SYSTEM — SMART IRRIGATION

15. System Overview

The soil moisture sensor and electrical actuator can be integrated into a fully automated irrigation system. The sensor continuously monitors soil moisture, and when levels fall below a set threshold, the Arduino sends a control signal to the actuator (via the motor driver) to open an irrigation valve or pump trigger, restoring adequate moisture levels.

16. System Block Diagram

Parameter	Specification
Input	Soil Moisture Sensor (Analog/Digital Output)
Controller	Arduino UNO / Mega (Processing & Decision Making)
Output Driver	L298N H-Bridge Motor Driver
Actuator	Electric Linear Actuator / Solenoid Valve
Power Source	5V (logic) + 12V DC (actuator)
Display/Monitor	LCD / Serial Monitor / IoT Dashboard (optional)

17. Operating Logic

Parameter	Specification
Sensor Reading > 700 (Dry)	Actuator Extends — Opens irrigation valve → Water flows
Sensor Reading < 300 (Wet)	Actuator Retracts — Closes irrigation valve → Water stops
300 < Reading < 700 (Optimal)	Actuator holds position — System in standby
Emergency Stop	Both IN1 and IN2 LOW — Actuator halts immediately

18. Conclusion

The soil moisture sensor and electrical actuator together form a powerful and efficient pair for automated, intelligent environmental control systems. The sensor provides critical feedback about soil conditions, while the actuator translates the controller's decision into physical action. When combined with a microcontroller platform like Arduino, this pair enables cost-effective, scalable, and energy-efficient automation for agriculture, research, and industrial applications.

The advancement of capacitive sensing technology and precision linear actuators has significantly improved reliability and accuracy. Future development includes IoT connectivity, AI-driven predictive control, and energy harvesting integration for fully autonomous field deployment.

19. References

- [Arduino Official Documentation — arduino.cc](#)
- [LastMinuteEngineers.com — Soil Moisture Sensor Tutorial](#)
- [Instructables — L298N Motor Driver Guide](#)
- [Texas Instruments — LM393 Datasheet](#)
- [Progressive Automations — Linear Actuator Technical Manual](#)
- [ResearchGate — Electric Actuator Circuit Analysis](#)
- [Rui Santos — Random Nerd Tutorials — Arduino Projects](#)